WORKED SOLUTION

STEP I 2007, Question 3

Pure • Trigonometric Integrals

STEP I 2007, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q3

The key idea

Neither $\cos^4\theta$ nor $\sin^4\theta$ is easy to integrate on its own, but their *difference* and their *sum* both are, once we use $\cos^2\theta + \sin^2\theta = 1$ to collapse them into functions of 2θ. Integrating the sum and the difference gives two simple linear equations for the two integrals we want. The same trick, one power higher, handles the sixth powers.

Solution

The identities

Factorising the difference of two squares,

$$\cos^4\theta - \sin^4\theta = (\cos^2\theta - \sin^2\theta)(\cos^2\theta + \sin^2\theta) = \cos 2\theta \cdot 1 = \cos 2\theta.$$

For the sum, we complete the square:

$$\begin{aligned}\cos^4\theta + \sin^4\theta &= (\cos^2\theta + \sin^2\theta)^2 - 2\sin^2\theta\cos^2\theta \\ &= 1 - \tfrac{1}{2}(2\sin\theta\cos\theta)^2 \\ &= 1 - \tfrac{1}{2}\sin^2 2\theta,\end{aligned}$$

using $\sin 2\theta = 2\sin\theta\cos\theta$. Both identities are proved.

The fourth powers

Write $C_4 = \int_0^{\pi/2}\cos^4\theta\,\mathrm{d}\theta$ and $S_4 = \int_0^{\pi/2}\sin^4\theta\,\mathrm{d}\theta$. Integrating the first identity,

$$C_4 - S_4 = \int_0^{\pi/2}\cos 2\theta\,\mathrm{d}\theta = \left[\tfrac{1}{2}\sin 2\theta\right]_0^{\pi/2} = 0.$$

For the second identity we need $\int_0^{\pi/2}\sin^2 2\theta\,\mathrm{d}\theta$. Using $\sin^2 2\theta = \frac{1}{2}(1-\cos 4\theta)$,

$$\int_0^{\pi/2}\sin^2 2\theta\,\mathrm{d}\theta = \left[\tfrac{1}{2}\theta - \tfrac{1}{8}\sin 4\theta\right]_0^{\pi/2} = \frac{\pi}{4},$$

since $\sin 2\pi = 0$. Hence

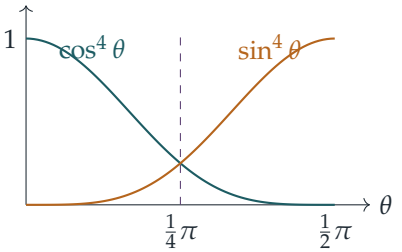
$$C_4 + S_4 = \int_0^{\pi/2} \left(1 - \tfrac{1}{2}\sin^2 2\theta\right) \mathrm{d}\theta = \frac{\pi}{2} - \frac{1}{2}\cdot\frac{\pi}{4} = \frac{3\pi}{8}.$$

Adding and subtracting the two equations,

$$\int_0^{\pi/2} \cos^4\theta \,\mathrm{d}\theta = \int_0^{\pi/2} \sin^4\theta \,\mathrm{d}\theta = \frac{3\pi}{16}.$$

Ask yourself Could we have seen that $C_4 = S_4$ without any calculation?

Yes. The substitution $\phi = \frac{1}{2}\pi - \theta$ turns $\cos\theta$ into $\sin\phi$ and swaps the limits 0 and $\frac{1}{2}\pi$, so $\int_0^{\pi/2} \cos^n\theta \,\mathrm{d}\theta = \int_0^{\pi/2} \sin^n\phi \,\mathrm{d}\phi$ for every n. The graphs of $\cos^4\theta$ and $\sin^4\theta$ are mirror images in the line $\theta = \frac{1}{4}\pi$, so they enclose equal areas:



So the first identity is really only needed for the sum; the difference integral being 0 is the symmetry in disguise.

The sixth powers

We repeat the method. With $c = \cos\theta$, $s = \sin\theta$ and $c^2 + s^2 = 1$:

$$\begin{aligned} c^6 - s^6 &= (c^2 - s^2)(c^4 + c^2s^2 + s^4) = \cos 2\theta\left(1 - \tfrac{1}{2}\sin^2 2\theta + \tfrac{1}{4}\sin^2 2\theta\right) \\ &= \cos 2\theta\left(1 - \tfrac{1}{4}\sin^2 2\theta\right), \end{aligned}$$

using the identities already proved and $c^2s^2 = \frac{1}{4}\sin^2 2\theta$. For the sum we use the cube of $c^2 + s^2$:

$$c^6 + s^6 = (c^2 + s^2)^3 - 3c^2s^2(c^2 + s^2) = 1 - \tfrac{3}{4}\sin^2 2\theta.$$

Write C_6 and S_6 for the two integrals. Since $\frac{\mathrm{d}}{\mathrm{d}\theta}\sin^3 2\theta = 6\sin^2 2\theta\cos 2\theta$,

$$C_6 - S_6 = \left[\tfrac{1}{2}\sin 2\theta - \tfrac{1}{24}\sin^3 2\theta\right]_0^{\pi/2} = 0,$$

as the symmetry argument also predicts. And, using $\int_0^{\pi/2}\sin^2 2\theta \,\mathrm{d}\theta = \frac{1}{4}\pi$ again,

$$C_6 + S_6 = \frac{\pi}{2} - \frac{3}{4}\cdot\frac{\pi}{4} = \frac{5\pi}{16}.$$

Therefore

$$\int_0^{\pi/2} \cos^6\theta \,\mathrm{d}\theta = \int_0^{\pi/2} \sin^6\theta \,\mathrm{d}\theta = \frac{5\pi}{32}.$$

Checks

- The answers decrease as the power rises: $\frac{1}{2}\pi$ for $n = 0$, $\frac{1}{4}\pi$ for $n = 2$, then $\frac{3}{16}\pi$ and $\frac{5}{32}\pi$, as they must since $0 \leq \cos\theta \leq 1$ on the interval.
- Numerically, Simpson's rule gives $\int_0^{\pi/2} \cos^4\theta \,\mathrm{d}\theta \approx 0.589049$ and $\int_0^{\pi/2} \cos^6\theta \,\mathrm{d}\theta \approx 0.490874$, matching $3\pi/16$ and $5\pi/32$; the sine integrals agree with these to the same accuracy.
- Putting $\theta = \frac{1}{4}\pi$ in the sum identities gives $\frac{1}{4} + \frac{1}{4} = 1 - \frac{1}{2}$ and $\frac{1}{8} + \frac{1}{8} = 1 - \frac{3}{4}$, both true.

Common slips

- Writing $\cos^4\theta + \sin^4\theta = 1$ by analogy with $\cos^2\theta + \sin^2\theta = 1$: the cross term $2\sin^2\theta\cos^2\theta$ is lost.
- Forgetting the factor $\frac{1}{2}$ from the chain rule when integrating $\cos 2\theta$ or $\cos 4\theta$; here it doesn't matter for $\cos 2\theta$ (the integral is 0 anyway) but it does for $\sin^2 2\theta$.

Beyond STEP

These are cases of Wallis's integrals $I_n = \int_0^{\pi/2} \sin^n\theta \,\mathrm{d}\theta$, which satisfy the reduction formula $I_n = \frac{n-1}{n} I_{n-2}$ (integrate by parts). You can check that $I_4 = \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$ and $I_6 = \frac{5}{6} I_4$ reproduce the answers above. Comparing I_{2n} and I_{2n+1} leads to Wallis's famous infinite product for π. Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice with reduction formulae, and Bartle and Sherbert, *Introduction to Real Analysis*, makes the limiting arguments rigorous.

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